

Dennis Liang

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EDUCATION

University of California, Berkeley

Master of Engineering in Visual Computing and Computer Graphics

August 2025 – May 2026

- MEng Fung Excellence Award

University of California, San Diego

Bachelor of Science in Computer Science, Minor in Business

September 2021 – March 2025

- Grades: 3.8/4.0 GPA, Provost Honors, Shores Scholarship
- Relevant Coursework: Computer Graphics II: Rendering, Computer Vision, Computer Animation, Machine Learning: Learning Algorithms, Advanced Software Engineering

TECHNICAL SKILLS

Programming Languages: Java, Python, C/C++, JavaScript, Typescript, HTML/CSS, Bootstrap, Markdown, SQLite, React, Sass, Redux, Smithy

Developer Tools: Atom, Git, GitHub, Visual Studio, VS Code, Vim, Jupyter Notebook, Android Studio, Figma

Concepts: Object-Oriented Programming, Software Design Patterns, and Algorithm, Agile Software Development, AWS, CI/CD

Software: Asana, Jira, GitHub Projects, Slack, Google Workspace, Microsoft Office, Wordpress

RELEVANT EXPERIENCE

Amazon

Seattle, WA

Software Development Engineer Intern | *AWS, Java, Typescript, React*

May 2025 – July 2025

- Designed and implemented a targeting service to segment sellers, aiming to reduce 45,000+ annual hours spent on manual segment list creation.
- Built and deployed scalable backend systems using AWS services including OpenSearch Service, DynamoDB, S3, ECS, and EC2.
- Collaborated cross-functionally to integrate the service into internal workflows while maintaining security and operational best practices.

Robolink

San Diego, CA

Software Engineer Intern | *Python, JavaScript, React, HTML, CSS*

June 2024 – February 2025

- Launched and enhanced Robolink's Web Python editor using React Pyodide: Identifying and resolving critical issues, improving functionality and user experience, contributing to a successful launch.
- Tested and implemented improvements for drone systems: Conducting comprehensive testing on drones, fixing software bugs, and developing a new dashboard system to streamline classroom learning management and enhance educational experiences.
- Streamlined the development pipeline from coding and testing to deploying new features, ensuring consistent performance improvements and maintaining high-quality standards across the software.

University of California, San Diego Center for Visual Computing

San Diego, CA

Undergraduate Research Assistant | *C++, Python, OpenCV, Spinnaker SDK*

January 2024 – June 2024

- Assisted in pioneering visual reconstruction research under a PhD student, focusing on the development and optimization of algorithms for enhanced imaging. Playing a key role in coding, debugging, and testing FLIR Blackfly S camera software, contributing to significant improvements in image quality and processing speed.
- Instrumental in setting up and calibrating advanced imaging equipment, ensuring optimal performance for experimental accuracy. Demonstrating technical proficiency and attention to detail in managing complex setups, leading to more efficient and accurate data collection for research objectives.

Early Transition in Tech Association

Remote

Software Engineer Intern | *Bootstrap, HTML, CSS, JavaScript*

September 2023 – February 2024

- Designed and developed a student-friendly web app learning platform using HTML, CSS, JavaScript, React, and Bootstrap to provide computer science education for under-served communities in Birmingham, AL
- Collaborated with an Agile software development startup team of 10 software engineers to create innovative technology solutions

PROJECTS

ZooSeeker

Team Lead | *Java, SQLite, Android Studio, Gradle*

April 2024 – June 2024

- Led and coordinated a team of developers to design, refactor, and enhance the Zoosseeker app, ensuring on-time delivery and adherence to high-quality standards.
- Managed user requirement analysis, task assignments, and project workflows, while spearheading improvements in functionality, performance, and user interface design.

Volumetric Path-Tracer

C++ Developer

April 2024 – June 2024

- Implemented a volumetric path tracer to render realistic visual effects such as subsurface scattering, volumetric shadows, and fog, achieving high visual fidelity.
- Developed and optimized physically-based rendering techniques to accurately simulate light interactions within participating media, showcasing expertise in ray tracing and volume rendering.

Ray-Tracer

C++ Developer

February 2024 – April 2024

- Designed and implemented a ray tracer to render photorealistic images, incorporating features such as reflections, refractions, and shadows, with rendered scenes including complex models like the Stanford dragon.
- Optimized rendering algorithms for efficiency, leveraging spatial data structures like bounding volume hierarchies (BVH) to enhance performance and reduce computation time.

Say-It AI Voice Assistant

Java Developer | *Java, OpenAI API, Whisper API, MongoDB, JUnit*

April 2023 – June 2023

- Designed and implemented a voice assistant for email and search functionality, ensuring seamless integration and enhanced user experience.
- Collaborated with a team of 5 software engineers in an Agile environment, conducting regular client feedback sessions to refine features and meet customer requirements.

OTHER EXPERIENCE

University of California, San Diego Cycling Team

San Diego, CA

Vice President

June 2024 – June 2025

- Leading weekly team rides across multiple cycling disciplines, fostering an inclusive environment for members of all skill levels and promoting team unity
- Coordinating with the executive board to organize events, manage logistics, and implement initiatives that enhance team growth, competition readiness, and member retention

University of California, San Diego Cycling Team

San Diego, CA

Media Liaison

May 2022 – June 2024

- Managed and created engaging content for the UC San Diego Cycling Team's Instagram, Facebook, and website, successfully promoting the team and increasing community engagement with an audience of over 1,000 followers.
- Collaborated with team officers and members to develop and execute media strategies, ensuring consistent and effective promotion of team activities, events, and accomplishments.

IEEE, University of California, San Diego Chapter

San Diego, CA

Outreach Chair

May 2022 – June 2023

- Planned and executed STEM outreach events for K-12 students in the San Diego community, inspiring interest in STEM fields and expanding educational opportunities.
- Recruited, organized, and led a team of university student volunteers to ensure successful event execution and impactful engagement with over 100 young participants.